Local stability does not imply global stability in the mixed-mechanism phosphorylation network

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Abstract. Conradi, Mincheva and Shiu (Bull. Math. Biol. 2019) studied the dual-site phosphorylation network with processive phosphorylation and distributive dephosphorylation, which has a unique steady state in every compatibility class and loses stability only through Hopf bifurcations. They asked whether a locally stable steady state is always globally stable in its class. It is not. For the integer rate constants $k = (1, 1, 1, 1, 300, 1, 3, 1, 1, 100)$ the Hopf bifurcation along an explicit line of compatibility classes is subcritical: we certify in ball arithmetic a simple pair $\pm i\omega_0$, the other four eigenvalues in the open left half-plane, a positive crossing speed and a positive first Lyapunov coefficient. Hence classes with a locally asymptotically stable steady state contain an unstable periodic orbit. At the rational class $(K_{\mathrm{tot}}, P_{\mathrm{tot}}, S_{\mathrm{tot}}) = (236/3,\ 6527/1800,\ 341699/5400)$ we prove, by exact arithmetic and a CAPD computer-assisted proof, that the steady state is locally asymptotically stable while the class contains an orbitally asymptotically stable periodic orbit and a second periodic orbit. The same conclusion holds on a nonempty open set of rate constants and total amounts. The conjecture of the same paper, that an unstable steady state always coexists with a stable periodic orbit, is not settled. Our example is consistent with it, and we certify one instance.

The question

Conradi, Mincheva and Shiu [1] (arXiv:1809.02886v2, Section 6, p. 20; journal version Section 6, p. 1849) end their study of the mixed-mechanism network with the question:

> *"Finally, we consider the dynamics in compatibility classes that contain a locally stable steady state. Our simulations suggest that such a steady state is in fact globally stable. Accordingly, we pose the question, Consider the mixed-mechanism network, and any choice of rate constants and total amounts. If the unique steady state x^* in $\mathcal{P}$ is locally stable, does it always follow that x^* is globally stable? In the Michaelis-Menten limit, this is true [36]."*

The same section (arXiv p. 18, journal p. 1848) states, as Conjecture 6.1 in both versions:

> *"Conjecture 6.1. Consider the mixed-mechanism network, and any choice of rate constants and total amounts. If the unique steady state in $\mathcal{P}$ is unstable, then $\mathcal{P}$ contains a stable periodic orbit."*

Conjecture 6.2 of [1] is a different statement (persistence of oscillations for large K_{tot}). The reference [36] in the quotation is Rao [2]. We answer the question in the negative (Theorems 4 to 6); Section 7 discusses what this implies for Conjecture 6.1.

1 Setting

The network. With $x_1, \dots, x_9$ the concentrations of S_0, K, S_0K, S_1K, S_2, P, S_2P, S_1, S_1P, the mass-action system (equation (2) of [1]) is $\dot{x} = g(x) = N\, r(x)$, where the

ten reactions and rates are

$$S_0 + K \xrightarrow{k_1} S_0K, \quad S_0K \xrightarrow{k_2} S_0 + K, \quad S_0K \xrightarrow{k_3} S_1K, \quad S_1K \xrightarrow{k_4} S_2 + K,$$
$$S_2 + P \xrightarrow{k_5} S_2P, \quad S_2P \xrightarrow{k_6} S_2 + P, \quad S_2P \xrightarrow{k_7} S_1 + P,$$
$$S_1 + P \xrightarrow{k_8} S_1P, \quad S_1P \xrightarrow{k_9} S_1 + P, \quad S_1P \xrightarrow{k_{10}} S_0 + P,$$

$r_1 = k_1x_1x_2$, $r_5 = k_5x_5x_6$, $r_8 = k_8x_6x_8$, and the other r_j are linear. The totals

$$K_{\mathrm{tot}} = x_2 + x_3 + x_4, \qquad P_{\mathrm{tot}} = x_6 + x_7 + x_9, \qquad S_{\mathrm{tot}} = x_1 + x_3 + x_4 + x_5 + x_7 + x_8 + x_9$$

are conserved; write $T = (K_{\mathrm{tot}}, P_{\mathrm{tot}}, S_{\mathrm{tot}}) = Wx$. The stoichiometric subspace $\mathcal{S} = \operatorname{im} N = \ker W$ has dimension 6, and the compatibility class of $T \in \mathbb{R}^3_{>0}$ is $\mathcal{P}_T = \{x \in \mathbb{R}^9_{\geq 0} : Wx = T\}$, a compact set that is forward invariant. By Prop. 3.1 [1] (from [3]), every $\mathcal{P}_T$ contains exactly one steady state, and it is positive.

Steady states. By Prop. 3.2 [1] the positive steady states are $\varphi(x_1, x_2, x_6)$ with $J = \frac{k_1k_3}{k_2+k_3}\, x_1x_2$ and

$$x_3 = \tfrac{k_1}{k_2+k_3}\, x_1x_2, \quad x_4 = \tfrac{J}{k_4}, \quad x_5 = \tfrac{(k_6+k_7)J}{k_5k_7x_6}, \quad x_7 = \tfrac{J}{k_7}, \quad x_8 = \tfrac{(k_9+k_{10})J}{k_8k_{10}x_6}, \quad x_9 = \tfrac{J}{k_{10}}.$$

Reduced coordinates. Put $z = (x_3, x_4, x_5, x_7, x_8, x_9) \in \mathbb{R}^6$. For fixed T, $x_1 = S_{\mathrm{tot}} - \sum_i z_i$, $x_2 = K_{\mathrm{tot}} - z_1 - z_2$, $x_6 = P_{\mathrm{tot}} - z_4 - z_6$, so that $x = \xi_T(z) = c_T + Ez$ with an integer 9×6 matrix E satisfying $WE = 0$, $PE = I_6$, where P selects the coordinates $3, 4, 5, 7, 8, 9$. Since $g(x) \in \mathcal{S}$ and every $s \in \mathcal{S}$ satisfies $s = EPs$, a function $x(t)$ with $Wx(0) = T$ solves $\dot{x} = g(x)$ if and only if $z(t) = Px(t)$ solves

$$\dot{z} = F_T(z) := P\, g(\xi_T(z)). \tag{1}$$

At a steady state x^* the Jacobian of (1) is $A = P\, Dg(x^*)\, E$, and $\det(\lambda I_9 - Dg(x^*)) = \lambda^3 \det(\lambda I_6 - A)$. We call x^* *locally asymptotically stable* (LAS) when all eigenvalues of A have negative real part, as in §2.2 [1], and *globally stable* in $\mathcal{P}_T$ when, in addition, every trajectory starting in $\mathcal{P}_T$ converges to x^*. A nonconstant periodic orbit in $\mathcal{P}_T$ therefore rules out global stability. Since g is quadratic, $D^2F_T(u, v) = B(u, v) := P\, D^2g(Eu, Ev)$ does not depend on z or T, and $D^3F_T = 0$.

The example. Throughout,

$$k^* = (k_1, \ldots, k_{10}) = (1, 1, 1, 1, 300, 1, 3, 1, 1, 100),$$

so $k_2 = k_6 = k_9$ (the setting of Prop. 4.1 [1]) and $k_{10}k_3k_4 - (k_3 + k_4)(k_3 + k_7)(k_4 + k_7) = 68 > 0$ (inequality (13) of [1]). We follow the line of steady states with $x_1 = 1/3$, $x_6 = 1/4$ and $x_2 = \mu > 0$:

$$x^*(\mu) = \varphi(\tfrac{1}{3}, \mu, \tfrac{1}{4}) = \Big(\tfrac{1}{3},\ \mu,\ \tfrac{\mu}{6},\ \tfrac{\mu}{6},\ \tfrac{2\mu}{675},\ \tfrac{1}{4},\ \tfrac{\mu}{18},\ \tfrac{101\mu}{150},\ \tfrac{\mu}{600}\Big), \tag{2}$$

$$T(\mu) = Wx^*(\mu) = \Big(\tfrac{4\mu}{3},\ \tfrac{1}{4} + \tfrac{103\mu}{1800},\ \tfrac{1}{3} + \tfrac{5761\mu}{5400}\Big). \tag{3}$$

$x^*(\mu)$ is the steady state of the class $\mathcal{P}_{T(\mu)}$. The matrix $A(\mu) = P\, Dg(x^*(\mu))\, E$ has rational entries, affine in μ; let $\lambda^6 + b_1(\mu)\lambda^5 + \cdots + b_6(\mu)$ be its characteristic polynomial and $H_i(\mu)$ its Hurwitz determinants ($i = 1, \ldots, 5$), polynomials in $\mathbb{Q}[\mu]$; H_5 has degree 9.

2 Results

Theorem 4 (subcritical Hopf bifurcation). There is $\mu_0 \in [a, b]$, where $a < b$ are the rationals stored in `certificates/hopf_point.json` ($b - a < 10^{-61}$, $\mu_0 = 59.16769751409043022823\ldots$), with the following properties.

(1) $A(\mu_0)$ has a simple pair of eigenvalues $\pm i\omega_0$, $\omega_0 = 2.24155834764\ldots$, and its four other eigenvalues have negative real parts.

(2) The eigenvalue $\lambda(\mu)$ continuing $i\omega_0$ satisfies $\operatorname{Re}\lambda'(\mu_0) > 0$.

(3) The first Lyapunov coefficient of $\dot z = F_{T(\mu_0)}(z)$ at $Px^*(\mu_0)$ is positive.

Consequently there is $\delta > 0$ such that for every $\mu \in (\mu_0 - \delta, \mu_0)$ the steady state $x^*(\mu)$ is locally asymptotically stable in $\mathcal{P}_{T(\mu)}$ and $\mathcal{P}_{T(\mu)}$ contains a nonconstant periodic orbit Γ_μ in $\mathbb{R}^9_{>0}$, which is unstable (one Floquet multiplier outside the unit circle, four inside) and shrinks to $x^*(\mu_0)$ as $\mu \uparrow \mu_0$. In particular $x^*(\mu)$ is not globally stable.

Theorem 5 (an explicit rational class). Let $k = k^*$ and

$$T = T(59) = \left(\tfrac{236}{3},\ \tfrac{6527}{1800},\ \tfrac{341699}{5400}\right).$$

The unique steady state of $\mathcal{P}_T$,

$$x^* = (\tfrac{1}{3},\ 59,\ \tfrac{59}{6},\ \tfrac{59}{6},\ \tfrac{118}{675},\ \tfrac{1}{4},\ \tfrac{59}{18},\ \tfrac{5959}{150},\ \tfrac{59}{600}),$$

is locally asymptotically stable. The class $\mathcal{P}_T$ contains two distinct periodic orbits in $\mathbb{R}^9_{>0}$: Γ_s, of period in $[2.96982298, 2.96982306]$, which is orbitally asymptotically stable, and Γ_u, of period in $[2.82910531, 2.82910541]$. Hence the set of initial conditions in $\mathcal{P}_T$ whose trajectories do not converge to x^* contains a nonempty relatively open set (the basin of Γ_s).

Theorem 6 (robustness). There is a nonempty open set $U \subset \mathbb{R}^{10}_{>0} \times \mathbb{R}^3_{>0}$ of rate constants and totals, containing $(k^*, T(59))$, such that for every $(k, T) \in U$ the steady state of $\mathcal{P}_T$ is locally asymptotically stable and $\mathcal{P}_T$ contains an orbitally asymptotically stable periodic orbit. Explicitly, $U = \{(k, W\varphi_k(y)) : k \in k^*(1 \pm \varepsilon),\ y \in (\tfrac{1}{3}, 59, \tfrac{1}{4})(1 \pm \varepsilon)\}$ (open boxes, componentwise) with $\varepsilon = 10^{-12}$.

Corollary 7. The answer to the question of §6 [1] is negative: there are rate constants and total amounts, forming a set with nonempty interior, for which the unique steady state in $\mathcal{P}$ is locally asymptotically stable but not globally stable.

3 Proof of Theorem 4

The family. For $\mu > 0$ consider $\dot z = \mathcal{F}(z, \mu) := P\, g(x^*(\mu) + Ez)$. This is (1) for the class $T(\mu)$ written in the shifted coordinate $z - Px^*(\mu)$, a polynomial family in (z, μ) with $\mathcal{F}(0, \mu) = 0$, $D_z\mathcal{F}(0, \mu) = A(\mu)$ and $D^2_z\mathcal{F}(0, \mu) = B$. A periodic orbit of $\dot z = \mathcal{F}(z, \mu)$ near 0 is mapped by $z \mapsto x^*(\mu) + Ez$ to a periodic orbit of $\dot x = g(x)$ in the affine space $\{Wx = T(\mu)\}$ near $x^*(\mu) > 0$, hence in $\mathcal{P}_{T(\mu)} \cap \mathbb{R}^9_{>0}$.

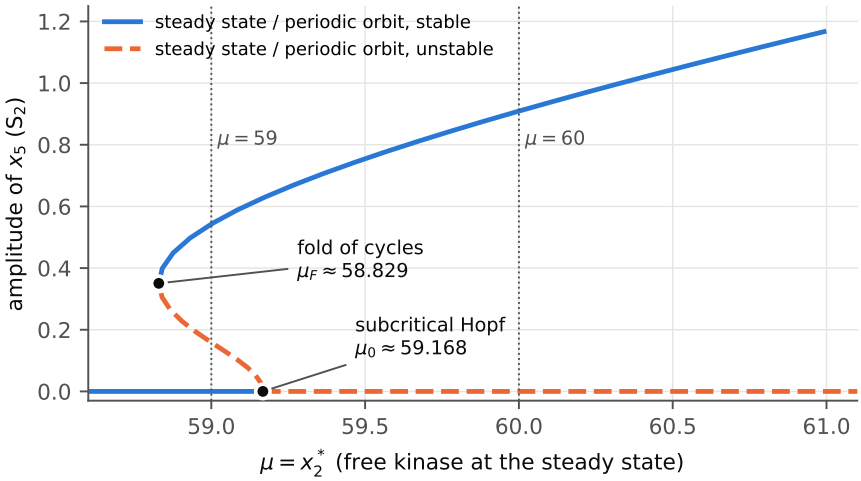


Figure 8 / Floating-point continuation of the periodic orbits along $\mu = x_2^*$ (rate constants k^*, $x_1^* = 1/3$, $x_6^* = 1/4$); amplitude of x_5 (doubly phosphorylated substrate). The unstable orbits born at the subcritical Hopf point μ_0 turn at a fold of cycles $\mu_F \approx 58.855$ into stable orbits that persist past μ_0. For $\mu_F < \mu < \mu_0$ a stable steady state coexists with a stable and an unstable orbit. Theorem 5 certifies the two orbits at $\mu = 59$; Proposition 10 certifies the stable orbit at $\mu = 60$. The figure itself is an illustration, not part of any proof.

Step 1: the bifurcation point (exact arithmetic). The certificate script of Section 8 computes $A(\mu)$, the coefficients $b_i \in \mathbb{Q}[\mu]$ and $H_1, \dots, H_5 \in \mathbb{Q}[\mu]$ exactly, checks the identity $\det(\lambda I_9 - Dg(x^*(\mu))) = \lambda^3 \det(\lambda I_6 - A(\mu))$ symbolically, and checks $H_5(a)H_5(b) < 0$ in exact rational arithmetic. By the intermediate value theorem $H_5(\mu_0) = 0$ for some $\mu_0 \in (a, b)$.

Step 2: the spectrum at μ_0 (ball arithmetic). Evaluating the exact polynomials on the ball $[a, b]$ with Arb [4] at 600 bits gives $H_1, \dots, H_4 > 0$, $b_6 > 0$ and $H_5' \neq 0$ on $[a, b]$. By the criterion of Liu [5] (see also Yang [6] and Prop. 2.3 [1]), $A(\mu_0)$ has exactly one pair of purely imaginary eigenvalues $\pm i\omega_0$, which is simple, and the other four eigenvalues have negative real parts; with $H_5'(\mu_0) \neq 0$ the pair crosses the imaginary axis transversally. As an independent check, ω_0^2 is a root of $b_1 s^2 - b_3 s + b_5$ (imaginary part of the characteristic polynomial at $i\omega_0$). Of its two roots, the enclosure of one makes the real part $-s^3 + b_2 s^2 - b_4 s + b_6$ contain 0 and the enclosure of the other excludes 0, which identifies $\omega_0^2 = s_0$. The cofactor $\lambda^4 + c_1\lambda^3 + c_2\lambda^2 + c_3\lambda + c_4 = p(\lambda)/(\lambda^2 + s_0)$, with $c_1 = b_1$, $c_2 = b_2 - s_0$, $c_3 = b_3 - s_0 c_1$, $c_4 = b_4 - s_0 c_2$, satisfies the four Routh–Hurwitz inequalities, and the remainders $b_5 - s_0 c_3$, $b_6 - s_0 c_4$ contain 0. In floating point the other four eigenvalues are approximately -4.17, -60.5, -88.8, -181.7.

Step 3: eigenvectors, Lyapunov coefficient and crossing speed (ball arithmetic). Let $M = A(\mu_0) - i\omega_0 I$, which has rank 5. The right eigenvector q normalised by $q_5 = 1$ is the unique solution of the 5×5 system obtained by deleting one row of $Mq = 0$ and moving the column of q_5 to the right-hand side. That subsystem is nonsingular (its ball Gaussian elimination succeeds), and the deleted equation is a consequence of the other five because $\operatorname{rank} M = 5$. Ball arithmetic therefore encloses q; the same construction for M^{T} encloses a left null vector w, and $\bar{p} := w/(w^{\mathsf{T}} q)$ satisfies $\bar{p}^{\mathsf{T}} A = i\omega_0 \bar{p}^{\mathsf{T}}$ and $\bar{p}^{\mathsf{T}} q = 1$. Kuznetsov's projection formula [7], §3.5 and Ch. 5 [8], with the cubic term absent, gives

the first Lyapunov coefficient

$$\ell_1 = \frac{1}{2\omega_0}\,\mathrm{Re}\Big[\bar{p}^{\mathsf{T}}\Big(-2B(q, A^{-1}B(q,\bar{q})) + B(\bar{q}, (2i\omega_0 I - A)^{-1}B(q,q))\Big)\Big], \tag{9}$$

with $A = A(\mu_0)$, which is invertible because $b_6 > 0$. Its sign does not depend on the normalisation of q (rescaling q by c multiplies ℓ_1 by $|c|^2$) nor on the choice of linear coordinates. The enclosure is

$$\ell_1 \in [5.86860833657 \times 10^{-4} \pm 10^{-17}] > 0.$$

For the simple eigenvalue $\lambda(\mu)$ with $\lambda(\mu_0) = i\omega_0$, $\lambda'(\mu_0) = \bar{p}^{\mathsf{T}} A'(\mu_0) q$, and the enclosure gives $\mathrm{Re}\,\lambda'(\mu_0) \in [0.00569698769645 \pm 3 \cdot 10^{-15}] > 0$. The linear solves in (9) are ball Gaussian eliminations, which enclose the exact solution for every matrix and right-hand side in the input balls; all inputs are balls containing the exact values at (μ_0, ω_0), so the output encloses the exact ℓ_1 and $\lambda'(\mu_0)$.

Step 4: the Hopf theorem. The family $\mathcal{F}$ is polynomial, $A(\mu_0)$ has the simple pair $\pm i\omega_0$ and no other eigenvalue on the imaginary axis, $\alpha'(\mu_0) > 0$ where $\alpha = \mathrm{Re}\,\lambda$, and $\ell_1 > 0$. In the notation of Hassard, Kazarinoff and Wan Ch. 1 [9], $\mathrm{Re}\,c_1(0) = \omega_0\ell_1 > 0$ (§3.5 [8]), so $\mu_2 = -\mathrm{Re}\,c_1(0)/\alpha'(\mu_0) < 0$ and $\beta_2 = 2\,\mathrm{Re}\,c_1(0) > 0$. The Hopf bifurcation theorem in $\mathbb{R}^n$ Ch. 1 [9], see also §§3.4–3.5, with the centre manifold reduction of Ch. 5 [8], gives $\delta > 0$ and, for every $\mu \in (\mu_0 - \delta, \mu_0)$, a unique nonconstant periodic orbit Γ_μ in a fixed neighbourhood of the origin. It shrinks to the origin as $\mu \uparrow \mu_0$, and its nontrivial characteristic exponent on the centre manifold is positive, so Γ_μ is unstable; the four other directions are contracting. For $\mu < \mu_0$ close to μ_0 the eigenvalue pair has negative real part (Step 3) and the other four eigenvalues remain in the open left half-plane by continuity, so $x^*(\mu)$ is LAS. Mapping back by $z \mapsto x^*(\mu) + Ez$ proves Theorem 4. □

4 Proof of Theorem 5

Local stability (exact). At $\mu = 59$, (2) gives the stated rational $x^* > 0$; $g(x^*) = 0$ and $Wx^* = T(59)$ hold exactly. The exact Hurwitz determinants of $A(59)$ are all positive ($H_5 = 4.398\ldots \times 10^{23}$) and $b_6 > 0$, so x^* is LAS by the Routh–Hurwitz criterion. Arb's certified eigenvalue enclosures of the exact rational matrix agree: six eigenvalues with negative real part, the closest to the axis being $-0.000956948\ldots \pm 2.24044\ldots i$.

The periodic orbits (CAPD). We use the reduced system (1) for $T = T(59)$, the Poincaré section $\Sigma = \{z_5 = c\}$ ($x_8 = c$, with c the double nearest to $5959/150$), crossed with x_8 increasing, and coordinates $y = (z_1, z_2, z_3, z_4, z_6)$ on Σ. The program `po_proof.cpp` uses the CAPD library [10] (rigorous Taylor integration with the C^1 Lohner algorithm [11]) to enclose the Poincaré map Π at a numerically computed point y_c and its derivative $D\Pi$ on a box $X \ni y_c$ of radius $10^{-8}(1 + |y_{c,i}|)$. With $C \approx (D\Pi(y_c) - I)^{-1}$ it checks

$$K(X) = y_c - C(\Pi(y_c) - y_c) + (I - C(D\Pi(X) - I))(X - y_c) \subset \mathrm{int}\, X.$$

By the Krawczyk theorem [12], §5.1 [13], Π then has a unique fixed point in X, that is, a periodic orbit of (1) crossing Σ in X. The rate constants and totals enter as intervals that contain the exact rationals, and the program also checks that all nine concentrations are positive on X. The checks pass for two boxes:

- $X_s \ni$ (8.4587632, 10.7368258, 0.6108895, 3.5188776, 0.0302965): return time in [2.969822982, 2.969823054], $\|I - C(D\Pi(X_s) - I)\|_\infty \le 0.067$;
- $X_u \ni$ (9.3018468, 10.2417384, 0.2773832, 3.3981275, 0.0643781): return time in [2.829105313, 2.829105405], $\|I - C(D\Pi(X_u) - I)\|_\infty \le 0.55$.

The two boxes are disjoint and each fixed point is a fixed point of the *first* return map, so each orbit crosses Σ upward only once per period; the two orbits are therefore distinct. Each orbit is positive at one point, hence everywhere, because $\mathbb{R}^9_{>0}$ is forward invariant and the orbit is periodic. Mapped by ξ_T, they are periodic orbits of $\dot{x} = g(x)$ in $\mathcal{P}_T$, and each return time is the minimal period, since the first return to Σ is the point itself.

Stability of Γ_s. With $S = \mathrm{diag}(1 + |y_{c,i}|)$, the program computes the interval matrix $(S^{-1}D\Pi(X_s)S)^{128}$ by repeated squaring and finds ∞-norm at most 0.296. Interval matrix multiplication is inclusion monotone, so for every matrix D in the enclosure $\rho(D) \le \|(S^{-1}DS)^{128}\|_\infty^{1/128} < 1$. In particular the derivative of Π at its fixed point, whose eigenvalues are the five nontrivial Floquet multipliers of Γ_s, has spectral radius below 1. Hence Π is a contraction near its fixed point y_s in a suitable norm, and $\Pi^n(y) \to y_s$ for every y in a neighbourhood $V \subset \Sigma$ of y_s. Every trajectory of (1) that starts close enough to Γ_s crosses Σ in V within a bounded time, by continuity of the flow and transversality of Γ_s to Σ. Continuity of the flow on bounded time intervals then shows that its distance to Γ_s tends to 0. So Γ_s is orbitally asymptotically stable within $\mathcal{P}_T$; by the Andronov–Witt theorem it even has an asymptotic phase (see e.g. [14]). Its basin is relatively open in $\mathcal{P}_T$, nonempty, and disjoint from the basin of x^*. □

In floating point, the nontrivial multipliers are about 0.9786 for Γ_s and 1.0044 for Γ_u, the next ones being of order 10^{-5}. So Γ_u is a saddle orbit, presumably the continuation of the Hopf orbits of Theorem 4 (Figure 8); neither fact is used in the proof.

5 Proof of Theorem 6

Let $\mathcal{B}$ be the closed box $k \in k^*(1 \pm \varepsilon)$, $y = (x_1, x_2, x_6) \in (\frac{1}{3}, 59, \frac{1}{4})(1 \pm \varepsilon)$, $\varepsilon = 10^{-12}$. The script `certify_open_set.py` checks:

(1) $\det D_y(W\varphi_k)(y) \in [1806.708 \pm 10^{-4}]$ on $\mathcal{B}$, so $(k, y) \mapsto (k, W\varphi_k(y))$ is a local diffeomorphism there, and the image U of the interior of $\mathcal{B}$ is open and nonempty;

(2) with an approximate eigenvector matrix R of A at the centre (exact double entries), the Gershgorin discs of the ball matrix $R^{-1}A(k, y)R$ (enclosing $R^{-1}A(k, y)R$ for every $(k, y) \in \mathcal{B}$) lie in $\mathrm{Re}\,\lambda \le -9.56 \times 10^{-4}$, so every steady state $\varphi_k(y)$, $(k, y) \in \mathcal{B}$, is LAS;

(3) the totals $W\varphi_k(y)$ over $\mathcal{B}$ lie in the boxes $K_{\mathrm{tot}} \in [78.666666667 \pm 5 \cdot 10^{-10}]$, $P_{\mathrm{tot}} \in [3.6261111111 \pm 4 \cdot 10^{-11}]$, $S_{\mathrm{tot}} \in [63.277592593 \pm 9 \cdot 10^{-10}]$;

(4) `po_proof_box.cpp`, the variant of `po_proof.cpp` in which the 13 parameters are additional constant state variables (so that the Lohner algorithm tracks their influence to first order), proves the Krawczyk inclusion and the spectral bound for Γ_s simultaneously for all rate constants in $k^*(1 \pm \varepsilon)$ and all totals in the boxes of item 3 ($\|I - C(D\Pi - I)\|_\infty \le 0.36$).

For $(k, T) \in U$, the steady state of $\mathcal{P}_T$ is $\varphi_k(y)$ for the corresponding y (uniqueness,

Prop. 3.1 [1]), which is LAS by item 2, and $\mathcal{P}_T$ contains an orbitally asymptotically stable periodic orbit by item 4 and the argument of Section 4. □

The width $\varepsilon = 10^{-12}$ reflects only the cost of the interval method (the fixed point moves by roughly $10^5\varepsilon$ while the Krawczyk box must stay small); the floating-point continuation of Figure 8 indicates a bistable window of relative width about 0.5% in μ alone.

6 How common is subcriticality?

The following observations are floating point and are not used in any proof. All Hopf points were located as sign changes of the largest real part of the spectrum along x_2, and ℓ_1 was evaluated with (9). That routine was validated against the planar Guckenheimer–Holmes formula and, for this model, by direct simulation at the Hopf point.

- For the rate constants of Table 1 [1], all 494 Hopf points found on a 50×50 grid of $(x_1, x_6) \in [10^{-4}, 10^4] \times [10^{-5}, 10^4]$ are supercritical, consistent with the Matcont observations of Figs. 2 and 4 [1]. The two Hopf points of their Figure 1 have $\ell_1 < 0$.
- With log-uniform random rate constants restricted to the region of Thm. 4.2 [1] (k_{10} above the bound (13), k_5/k_8 large), 15 929 of 70 884 Hopf points (22%) are subcritical; with unrestricted log-uniform rate constants on $[10^{-2}, 10^2]$, 44 of 1 259.
- For k^* itself, about 8% of the Hopf points found on a grid of (x_1, x_6) are subcritical, so ℓ_1 changes sign on the Hopf surface: there are generalised Hopf (Bautin) points.

Subcritical Hopf points are therefore not exceptional, but they are absent near the parameter values used in [1], which explains the simulations reported there.

7 Conjecture 6.1

Theorems 4 and 5 do not contradict Conjecture 6.1: they concern classes with a stable steady state. They do show that the stable/unstable dichotomy of the steady state is not a dichotomy between convergence and oscillation, because oscillations also occur in classes with a stable steady state. In our example, the unstable orbits born at the Hopf point fold into stable orbits that persist on the unstable side (Figure 8), as Conjecture 6.1 predicts. We certify one instance.

Proposition 10. For $k = k^*$ and $T = T(60) = (80,\ 221/60,\ 5791/90)$ the steady state $x^* = (\frac{1}{3}, 60, 10, 10, \frac{8}{45}, \frac{1}{4}, \frac{10}{3}, \frac{202}{5}, \frac{1}{10})$ is unstable (exactly two eigenvalues with positive real part, $0.0047032\ldots \pm 2.24702\ldots i$), and $\mathcal{P}_T$ contains an orbitally asymptotically stable periodic orbit (return time in [3.09916969, 3.09916978]).

Proof. Exact Hurwitz determinants ($H_1, \ldots, H_4 > 0$, $H_5 < 0$, $b_6 > 0$), certified Arb eigenvalue enclosures, and the CAPD proof of Section 4 with $\mu = 60$ (`certify_orbits.py`). □

Two remarks on what a proof of Conjecture 6.1 would need. First, when $k_2 = k_6 = k_9$, Prop. 4.1 [1] gives $H_1, \ldots, H_4 > 0$ and $b_6 > 0$ at every steady state. By Routh's theorem Ch. XV [15], a steady state that is not LAS and not on the Hopf surface ($H_5 < 0$) then has exactly two eigenvalues in the open right half-plane and none on the axis, hence a

two-dimensional unstable manifold in its class; in 3×10^5 random floating-point samples, with unequal dissociation constants allowed, we never observed more than two. Second, the conjecture is a global statement about all classes with an unstable steady state, including those far from any Hopf point. Our methods certify individual orbits but give no handle on, for instance, excluding a stable invariant torus or a chaotic attractor without stable periodic orbits. We did not search systematically for such attractors. Conjecture 6.1 remains open.

8 Computer verification

All scripts are in `scripts/` and print PASS or FAIL for every claim, with a non-zero exit code on failure (Python 3.12, SymPy 1.14, python-flint 0.9; CAPD 6.1 built from https://github.com/CAPDGroup/CAPD in WSL Fedora).

- `certify_subcritical_hopf.py` (8 s): claims C0–C8 of Theorem 4: conservation laws; the parametrisation (symbolic in all k_i and x_1, x_2, x_6); the reduction identity $\lambda^3 p_A$; the exact sign change of H_5; ball checks of $H_1, \dots, H_4$, b_6, H_5'; identification of ω_0 and the Hurwitz cofactor; $\ell_1 > 0$; $\operatorname{Re}\lambda' > 0$; the totals (3). Data: `certificates/hopf_point.json`, generated by `make_hopf_point.py`.
- `certify_orbits.py` (about 40 s, mostly compilation): exact steady states, Hurwitz determinants and Arb eigenvalue enclosures at $\mu = 59, 60$; a symbolic check that the vector field typed in `po_proof.cpp` equals $P\,g(\xi_T(z))$ for symbolic k and T; compilation and runs of `po_proof` for the three orbits, with rationals enclosed by outward-rounded doubles. Data: `certificates/orbits.json`; log: `certificates/po_proof_log.txt`.
- `certify_open_set.py` (about 1 min): items 1–4 of Section 5; its log is `certificates/open_set_log.txt`.
- `exploration/`: the floating-point searches, continuation and figure (not certificates).

Assumptions and limitations. The proofs rely on: the uniqueness of the steady state in each class Prop. 3.1 [1] (only for the word "unique" and in Theorem 6); the Hopf theorem with the stability formula of [9] and the projection formula (9) [7] [8]; Liu's criterion [5]; the Krawczyk theorem; and the correctness of the Arb, FLINT and CAPD implementations of interval arithmetic and rigorous integration. Theorem 4 gives no explicit δ; Theorem 5 is the quantitative counterpart. The value μ_F of the fold of cycles, the amplitudes in Figure 8, the statements of Section 6 and the identification of Γ_u with the Hopf orbits are numerical and not proved.

Related work. We found no computation of Lyapunov coefficients for this network. Vassena [16] and Feliu and Kaihnsa [17] concern the fully distributive network. The Evidence Press record [18] treats reduction conjectures of [17] and states that it does not resolve the dual-phosphorylation question.

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